Lab Report: Image classification experiments

1 Introduction

In this report we compare several models on standard benchmarks. The dataset was split into 70 percent training data and the rest for testing. We trained each model for 30 epochs with a batch size of 128.

2 Results

Figure 1 shows the validation accuracy by dataset.

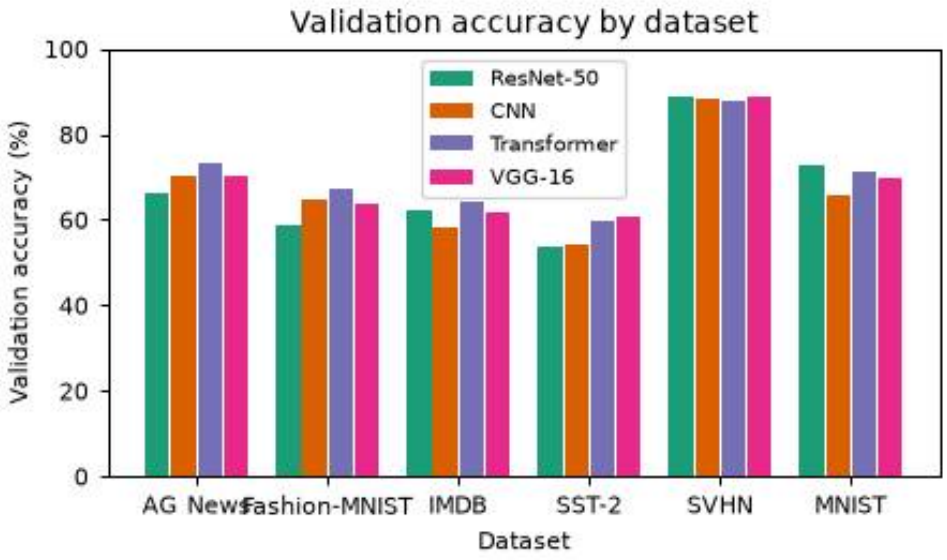


Figure 1: Validation accuracy by dataset for the models in this experiment.

Evaluated on SVHN, ResNet-50 comes in at 89% validation accuracy (Figure 1). As shown in Figure 1, Evaluated on AG News, ResNet-50 comes in at 116.1% validation accuracy. Evaluated on SST-2, ResNet-50 comes in at 53.5% validation accuracy (Figure 1). ResNet-50 reaches 43.5% validation accuracy on SST-2.

Figure 2 shows the accuracy of ResNet-50 by dataset.

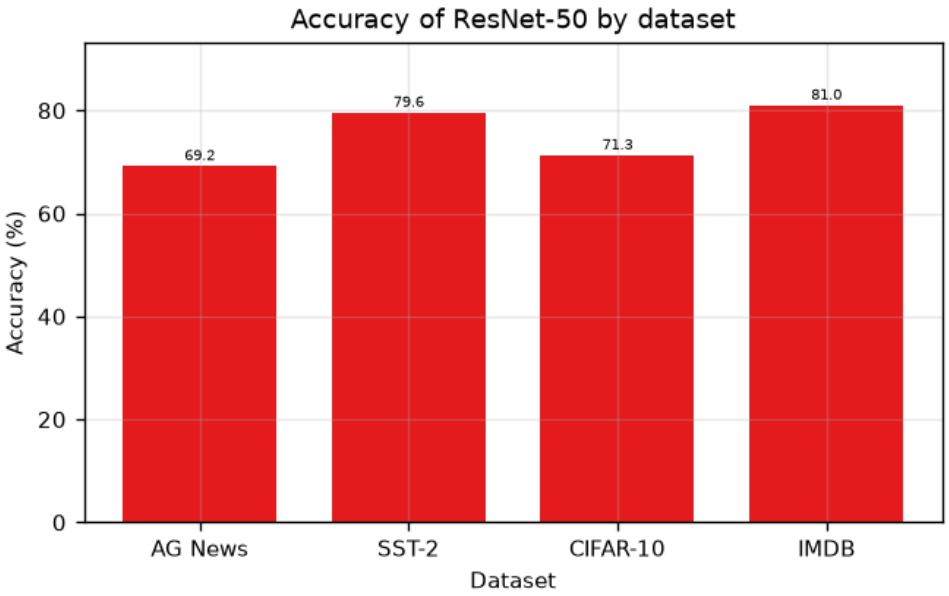


Figure 2: Accuracy of ResNet-50 by dataset for the models in this experiment.

ResNet-50 scores 10.7 points higher on IMDB than on SST-2. ResNet-50 reaches 79.6% accuracy on SST-2. Evaluated on SST-2, ResNet-50 comes in at 88.9% accuracy. ResNet-50 achieves 127.6% accuracy on IMDB (Figure 2).

Figure 3 shows the test accuracy during training.

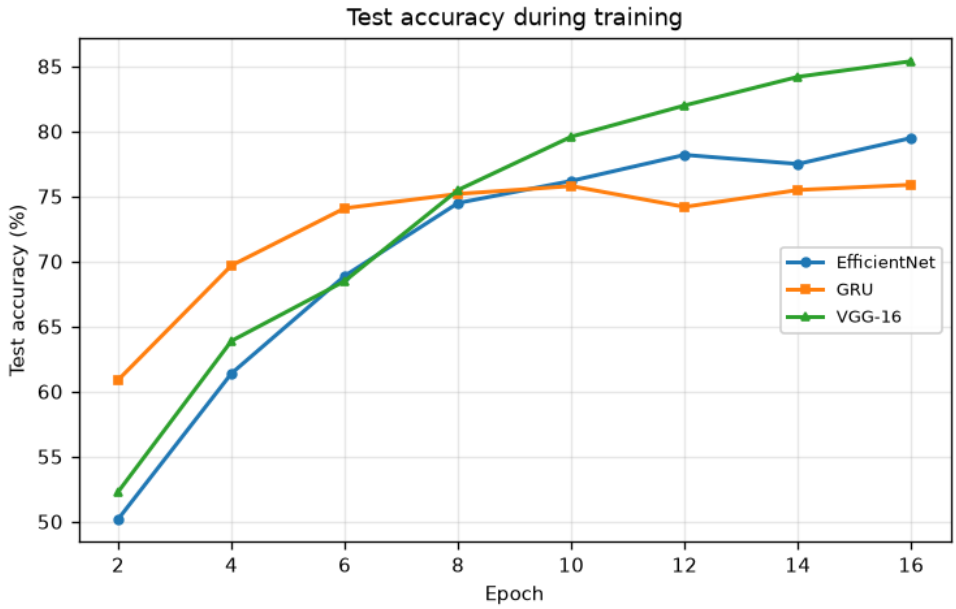


Figure 3: Test accuracy during training for the models in this experiment.

From epoch 6 to epoch 10, the test accuracy of GRU improves by a relative 4.9%. At epoch 16, EfficientNet is 4 points better than GRU. Epoch 2 sees VGG-16 post a test accuracy of over 50% (Figure 3). As shown in Figure 3, GRU reaches 75.0% test accuracy at epoch 12.

3 Discussion

Random Forest reaches 61.1% accuracy on MNIST. We note that the training was stable for all models.